

# Hum 4147
# Technology, Environment and Society

**Faruque Ahmed Haolader, PhD**
Professor, TVE Department, IUT

1

## Content

- Sustainability and Sustainable Development
- Environmental Stewardship
- Environmental Justice and Equity



2

Hum 4147: Technology, Environment and Society

## Sustainability and Sustainable Development



3

Hum 4147: Technology, Environment and Society

## Sustainability and Sustainable Development

In 1987, the United Nations World Commission on Environment and Development released the report Our Common Future, commonly called the Brundtland Report.[1] The report included a definition of "sustainable development" which is now widely used:[1][10]

**Sustainable development is a development that meets the needs of the present without compromising the ability of future generations to meet their own needs.**

**It contains two key concepts within it:**

https://en.wikipedia.org/wiki/Sustainable_development

4

Hum 4147: Technology, Environment and Society

# Sustainability and Sustainable Development

The concept of '**needs**', in particular,
the essential needs of the world's poor, to which overriding priority should be given; and

The idea of **limitations** imposed by
the state of technology and social organization on the environment's ability to meet present and future needs.

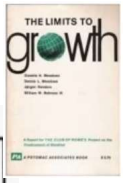
— World Commission on Environment and Development, Our Common Future (1987)

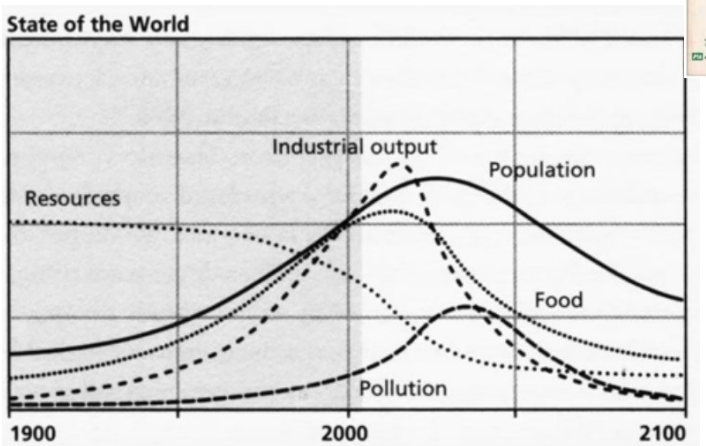
https://en.wikipedia.org/wiki/Sustainable_development

5

## **Sustainability** and Sustainable Development

Need for sustainable development issue came after team of scientist from MIT, 1972




Donella H. Meadows, Dessis L. Meadows, Jorgen Randers, and William W. Behrens III

https://www.un.org/en/desa/state-our-world



6

## Sustainability and **Sustainable Development**

"Sustainable development meets the needs of the present without compromising the ability of future generations to meet their own need"

The World Commission on Environment and Development, United Nations, 1987

Arises many questions.....what will we need?

We do know that we are using resources very fast

7

## Sustainability and Sustainable Development

**Sustainability**

Stocks and Flow → Exp.; expenses ≤ income

**Development**

Well-being → Can mean different things to different people

Comfort, happiness, health

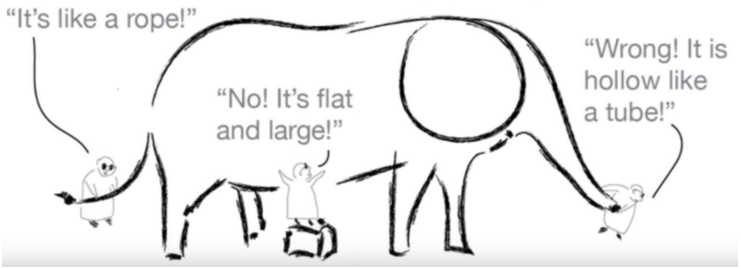
Inhere rent Challenge



8

## Sustainability and Sustainable Development

### The Parable of the Elephant and 3 Blind Men




Multiple perspectives are required to understand

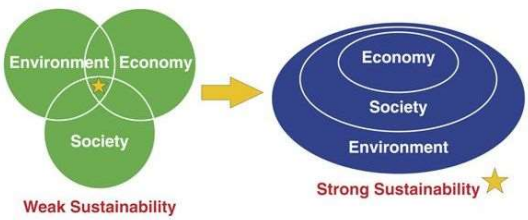


9

## **Sustainability** and Sustainable Development

Often sustainability viewed as a balance between
**Economy**, **Environment** and **Society** – **traditional model**

**False Notion**
-Economic activity outside the environment
- inherent competition
- separate goals




**Alternative Model**
- One environment
- society develops inside the env.
- Economic activity

Source: Maureen Hart - Sustainable Measures



10

## Sustainability and **Sustainable Development**

### 5 Important Measures for Sustainable Development

**(i) Technology:**

Using appropriate technology is one which is locally adaptable, eco-friendly, resource efficient and culturally suitable.



This concept is known as "design with nature"

http://www.yourarticlelibrary.com



11

## Sustainability and Sustainable Development

### 5 Important Measures for Sustainable Development

**(ii) Reduce, Reuse, and Recycle Approach:**

The 3-R approach advocating minimization of resource use, using them again and again instead of passing it on to the waste stream.

**(iii) Promoting Environmental Education and Awareness:**

Making environmental education the centre of all learning process will greatly help in changing the thinking pattern and attitude of people towards our earth and the environment.

feeling of belongingness to earth

'Earth thinking'

http://www.yourarticlelibrary.com



12

Sustainability and Sustainable Development

## 5 Important Measures for Sustainable Development

**(iv) Resource Utilization as per Carrying Capacity:**
If the carrying capacity of a system is crossed (say, by over exploitation of a resource), environmental degradation starts and continues till it reaches a point of no return.

**Carrying capacity has two basic components:**
i. Supporting capacity i.e. the capacity to regenerate
ii. Assimilative capacity i.e. the capacity to tolerate different stresses.

**(v) Improving Quality of Life Including Social, Cultural and Economic Dimensions:**
Ensuring the distribution of fare share of benefits

http://www.yourarticlelibrary.com



13

Hum 4147: Technology, Environment and Society

# Sustainability and Sustainable Development

Environmental Stewardship

Environmental Stewardship

14

Environmental Stewardship

## Introducing Environmental Stewardship

Stewardship of the environment refers to **protecting the environment** through **recycling, conservation, regeneration, and restoration**.

It means **taking responsibility for our choices**.
The responsibility for environmental quality should be shared by all those whose actions affect the environment.



15

Environmental Stewardship





16

## Environmental Stewardship

**Defining Environmental Stewardship**

Individual or Organizations practice:

✓ Use resources efficiently

✓ Reduce waste and minimize on the negative impacts on the environment

✓ Translate reduced consumption and waste into financial savings

✓ Provide a healthy, more efficient and effective working environment



17

## Environmental Stewardship

**Defining Environmental Stewardship**

Embodies the concept of a Green Office, a Green School or Green Home which is conceptualized as smarter and better as it is:

❑ Ecological - using non-toxic, recycled, environmentally friendly products and supplies

❑ Efficient – using as little energy and other resources as possible and putting out the smallest amount of waste as possible

❑ Healthy – will generate as little visual, noise and physical pollution as possible



18

## Environmental Stewardship

### Principles of Environmental Stewardship

**Core Principle 1: Resource Conservation** – making the most efficient use of resources (materials, energy and water) through:

- Energy Efficiency and Conservation
- Water Efficiency and Conservation
- Solid Waste Management and Conceptualizing Waste as a Resource

**Core Principle 2: Pollution Prevention** – minimizing contamination of the environment by chemicals or other materials through:

- Vehicle/Fleet Management
- Management of the Use of Chemicals and Other Hazardous Substances

**Core Principle 3: Occupational Health and Safety**

- Managing Indoor Air Quality
- Comfort and Productivity in the Workplace
- Emergency and Disaster Management



19

## Environmental Stewardship

### Benefits of Implementing an Environmental Stewardship

✓Economic Benefits Human Health

✓Health Benefits

✓Protection and Conservation of the Environment





20

Hum 4147: Technology, Environment and Society

## Environmental Justice and Equity



21

Hum 4147: Technology, Environment and Society

## Environmental Justice and Equity

.. is the **fair treatment** and **meaningful involvement** of all people regardless of race, color, national origin, or income with respect to the development, implementation, and enforcement of environmental laws, regulations, and policies.



22

Hum 4147: Technology, Environment and Society

# Environmental Justice and Equity

**Fair treatment means**

….that no group of people should bear a disproportionate share of the negative environmental consequences resulting from industrial, governmental and commercial operations or policies.



23

Hum 4147: Technology, Environment and Society

# Environmental Justice and Equity

**Meaningful involvement means**

- people have an opportunity to participate in decisions about activities that may affect their environment and/or health;
- the public contribution can influence the regulatory agency's decision;
- their concerns will be considered in the decision making process; and
- the decision makers seek out and facilitate the involvement of those potentially affected.



24

# Environmental Justice and Equity

**What is the Problem?**

- The impact of environmental policies and decisions have differentiated impacts across population groups
- Most countries have utterly no idea what environmental justice is nor have any mechanism for addressing EJ issues

**Examples**

- Haiti and much of the developed Western Hemisphere
- Environmental neglect in New Orleans
- The former East German Republic and its neighbors
- Philippines Islands and Japan
- Sub-Saharan Africa and Europe



25



# Environmental Justice and Equity

**National Policy Alternatives**

- Improving information exchange? Knowledge is power and leverage
- National or state environmental justice legislation or regulation
- Non-governmental action groups or agenda setting

**Outcomes and Futures**

- EJ resolution cannot solve all the world's social/economic justice problems
- International EJ issues may be the next great world policy issue



26

Hum 4147: Technology, Environment and Society

Climate and Climate Change



27




28

## Climate and Climate Change

**Climate** – is the statistics of the environment over time. It can include seasonal cycle, extreme events, year to year variations

Overall weather pattern of precipitation and air temperature in an specific area over time.

**Weather** – what is happening right now in the atmosphere. It's a combination of Temperature, Humidity, Precipitation, Cloudiness, Visibility, Wind and Atmospheric pressure.

Or

Weather – is the condition of the outside air or atmosphere at a particular time and place

29

## Climate and Climate Change

**Climate change** refers to any **significant change** in measures of climate (such as temperature, precipitation, or wind) lasting for an extended period (decades or longer).

- Climate change may result from:
  - natural factors, such as changes in the sun's intensity or slow changes in the Earth's orbit around the sun;
  - natural processes within the climate system (e.g. changes in ocean circulation);
  - human activities that change the atmosphere's composition (e.g. through burning fossil fuels) and the land surface (e.g. deforestation, reforestation, urbanization, desertification, etc.)

30



## Climate and Climate Change

### Seven Theories of Climate Change

1. **Anthropocentric Global Warning Theory**
2. Bio-thermostat Theory of Climate Change
3. Cloud Formation and Albedo
4. **Human Forcing Besides Greenhouse Gases**
5. Ocean Current
6. Planetary Motion
7. Solar Variability



31

**Thanks**
**floor is open for Q& A**

32